AGI ALPHA FLYWHEEL

Making the Next Opportunity Possible

Specialist Intelligence and the Expanding Frontier of Profitable Work

An executable study of numerical learning, certified delivery and selective investment

Vincent Boucher
President of MONTREAL.AI

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The result worth acting on. A learned numerical specialist transferred successfully and accelerated fresh design-selection work. Competent ordinary tools already delivered the same service. Additional Gym practice increased downstream verification work, while modeled specialist investment failed to repay its complete additional cost at the tested workloads.

TRANSFORMATIONAL INTELLIGENCE ∞

LEARNING TO CHANGE THE GAME

FIND YOUR ALPHA. BUILD YOUR SUCCESSOR Ω.

Observed numerical execution; sourced model specifications; explicitly assumed commercial conditions. No customer sales, physical control trials, live ENS admission or measured frontier-model task performance are claimed.

Abstract

Can an automated enterprise repeatedly expand the set of work it can profitably deliver by learning specialists, transferring their capabilities and discovering adjacent offerings? We operationalize this question as a conjunction of usefulness, accuracy, reliability, deadline, customer-effort and complete-cost requirements. A faster component is insufficient: a specialist must change whether a fixed offering qualifies, or improve the economics of work already feasible.

We execute a bounded study of stationary disturbance-energy analysis for synthetic dissipative dynamical systems. Six originating examples produce a covariance-weighted reduced basis; an optional Gym adds four development examples. On 64 protected confirmation cases, rank-48 work and Gym bases each certify 62 cases, while ordinary adaptive Krylov certifies all 64. All learned bases fail certification on 32 stronger-coupling cases, demonstrating the need for refusal and fallback. A separate process reuses the digest-verified artifact without training snapshots. On a final fresh eight-case numerical workflow, ordinary adaptive delivery takes 0.4662 seconds, work-derived reuse 0.1728 seconds and Gym-derived reuse 0.4592 seconds. Every policy returns the direct oracle's discrete-grid optimum. The ordinary baseline already meets the frozen two-second numerical contract; no uniquely enabled offering is demonstrated.

A paired enterprise simulation connects these measured technical costs to explicit pricing, demand, capacity, failed commercialization, collections, obligations and optional learning assumptions. Existing AGI ALPHA accounting and trust code is exercised without alteration. The central contribution is an inspectable investment boundary and a defensible stopping policy: preserve profitable ordinary delivery when its economics hold, retain only advantages with a credible repayment mechanism, and withhold claims of new markets or sustained compounding until additional evidence exists.

1 What this study establishes

The motivating opportunity is substantial in principle: productive experience can leave behind a reusable capability; that capability can reduce the difficulty of a subsequent mission; new missions can generate further useful experience. Each arrow, however, has a separate empirical burden. This study follows those arrows through executable numerical work and an explicit commercial model. It deliberately permits a competent ordinary enterprise to succeed and to decline poor investments.

The principal measured result is *improved delivery of an already feasible service*. Work-derived reuse is approximately 2.70 times faster than the adaptive ordinary comparator on the final fresh design-selection batch. That ratio is real within the recorded machine, source and synthetic problem family. Its absolute saving is 0.2934 seconds for an eight-case service. Neither the ratio nor its absolute value establishes customer willingness to pay, physical-model adequacy or a new commercially feasible offering.

The optional Gym campaign changes the retained artifact but does not improve protected acceptance counts. On the downstream selection problem it requires additional direct-solver refinements, removing most of the work basis's advantage. This is a useful example of negative transfer at the service level: a representation with favorable primitive residual behavior can impose more coordination and verification work on the next task.

Claim	Evidence in this edition	Supported conclusion
Actual specialist learning	Data-dependent basis construction and Gym refit; versioned arrays	A numerical representation was learned
Fresh technical reuse	Separate recipient process; protected and fresh numerical cases	An executable artifact transfers locally
Newly feasible service	Strong ordinary comparators meet standard analysis and design contracts	No unique specialist crossing established
New offering discovery	Same finite proposal grammar; fresh adjacent prototype; unsupported monitor rejected	Bounded prototype evaluation, not open-ended invention
Profitable expansion	Paired simulation with costs, failures and alternatives	Conditional policy/economic results only
Authenticated network	Native local signer/revocation tests; production scope unchanged	No live ENS admission or network deployment
Sustained compounding	No repeated customer-validated capability-created service chain	Not established

Table 1: Claims are kept at the level of their evidence. "Not established" does not mean impossible; it specifies the next evidentiary obligation.

2 A productive frontier with meaningful boundaries

Let an offering o specify a beneficiary, useful outcome, deliverable, input contract, required quality q_o, reliability r_o, deadline d_o, maximum customer effort e_o, and price p_o. For policy π, environment z and workload N, define qualification as

$$F(o, \pi, z, N) = \mathbf{1}\{U_o,\ Q_{\pi,o} \geq q_o,\ R_{\pi,o} \geq r_o,\ T_{\pi,o} \leq d_o,\ E_{\pi,o} \leq e_o,\ \Pi_{\pi,o}(N) \geq 0\}. \tag{1}$$

U_o represents a useful outcome for the stated beneficiary. The profit term includes a defensible share of development, validation and lifetime obligations. Each predicate requires evidence; an internal model score cannot substitute for it. The productive frontier is the set of offerings for which these joint requirements hold under the declared conditions.

We distinguish four possible outcomes. *Improved delivery* increases quality or surplus inside an existing feasible set. *Economic enablement* makes technically possible work affordable or dependable enough to offer. *Compositional enablement* joins capabilities whose combined operation crosses a pre-existing requirement. *Opportunity discovery* identifies and validates a useful offering that was not merely named in advance. These categories are cumulative evidentiary claims, not interchangeable marketing descriptions.

The comparison is always against the best credible alternative within the declared search. A dense solve is an important correctness reference, but an enterprise with ordinary adaptive methods is the relevant commercial control. We therefore report dense-solver threshold misses without interpreting them as a specialist-created frontier when an ordinary adaptive method already qualifies.

There is also a distinction between a finite *research gate* and an operational reliability claim. A 62/64 pass rate exceeds a 95% empirical threshold. It does not establish a population reliability of at least 95%. Similarly, two measured batches cannot establish a dependable latency service-level agreement. The full commercial frontier remains conditional on untested customer effort, physical input adequacy and market behavior even when the numerical research gate passes.

3 Opportunity portfolio and commercial mechanism

We examined several technically demanding opportunities before selecting one bounded portfolio. The selection considered verifiable useful outcomes, credible bottlenecks, rights-cleared artifacts, ordinary alternatives and overlap with existing project owners. It did not select a domain solely because favorable numerical results were available.

Candidate	Beneficiary and bottleneck	Verifiable deliverable	Selection / falsifier
Control-model analysis	Engineering teams comparing disturbance response across candidate models	Covariance/energy report, numerical certificate, bounded selection result	Selected; ordinary Krylov may already meet every requirement
Thermal parameter sweeps	Teams repeatedly exploring a validated geometry	Certified parameter-response table and reproducible solver	Credible, but already owned by an active research lane
Manufacturing disruption planning	Operators repairing resource-constrained schedules	Feasible schedule, constraint proof and objective gap	Credible, but separately owned; demand and operational integration remain burdens
Scientific software assurance	Research groups needing reproducible numerical pipelines	Correctness checks, portability and regression evidence	Useful; analyst/data qualification may dominate runtime
Algorithm / experiment invention	Teams seeking a better search or information-gathering policy	Executable algorithm and fresh performance comparison	High potential, but broader search and buyer validation exceed this edition

Table 2: A focused portfolio follows technical and commercial mechanisms. The companion retains sources and rejected-candidate reasoning.

3.1 The first deliverable

The proposed first service is an offline, reproducible disturbance-energy analysis of already validated linear models. A customer supplies rights-cleared matrices, units, disturbance assumptions and a precise question. The output includes validated inputs, a covariance or reproducible representation, an energy estimate, an explicit residual-derived numerical error bound, refusal messages where assumptions fail, and a machine-readable evidence package. This is a scientific-computation deliverable. It does not certify a physical installation, authorize a controller or replace professional engineering obligations.

The selected numerical computation has an established role. Continuous-time Lyapunov equations underlie stationary covariance and $\mathcal{H}_2$ disturbance-response calculations. Existing software already includes dense and large-scale low-rank methods; M.E.S.S. and integrated Krylov–ADI work are relevant acquisition alternatives, not missing capabilities that an AI enterprise may assume away [1, 2, 3, 4].

Two adjacent offerings are considered. A standard batch report compares 32 model variants. A finite-grid design report selects added damping while preserving the disturbance-energy objective and numerical requirements. The frozen interactive 0.25-second tier is assessed separately and remains unqualified as a reliable service. The commercial simulation uses the distinct standard five-second batch contract already defined in Experiment A; it does not relabel the interactive failure as success.

3.2 What the market evidence can support

No exact matched market quote, customer interview, signed purchase commitment or actual sale was established for this narrow offline covariance report. Related control consulting is advertised, and official wage data provide a labor-cost reference, but neither is a price for the tested synthetic artifact. The BLS source reports electrical-engineering wages; a conversion from annual wages to hourly wages and then to a consultancy billing rate requires explicit additional assumptions [5, 6].

The economic model uses estimated equivalent human effort of two, four and eight hours at an assumed USD150 per hour, producing reference anchors of USD300, USD600 and USD1,200. These are transparent scope-and-rate estimates. They are not documented market prices. Buyer value, trust, acquisition cost and workload are separate hypotheses. The finite price search may select a substantial discount, but neither the discount nor a low model bill guarantees a purchase.

A continuous-monitoring service is considered as a genuinely different buyer-value hypothesis without inventing a human equivalent. It is declined because streaming assurance, a working prototype, demand evidence and an operational authority boundary are absent. The cost of assessment is retained in the declared research/production setup budget. A new name does not qualify as a new market.

4 System reuse, source identity and authority

The implementation starts from AGI ALPHA source `80b345828f34fd8a6e592977dd17ef37141305f4`. The coordination thread was inspected and a bounded additive study path reserved. Active thermal/invention and manufacturing research owners were preserved. All new work is under `studies/agi-alpha-productive-frontier-v1`; no production runtime, identity policy, settlement boundary or earlier paper is overwritten.

The study directly executes unchanged native `Journal`, `TrustRegistry`, domain-separated signing and `CostLedger` interfaces. A 13-file source closure is checked against exact SHA-256 and Git blob identities before import. Directed tests cover replay identity, duplicate costs across buckets, incomplete records, unknown costs, revoked signers, mutated payloads, unpaid invoices, partial payments and source drift. A successful signature authenticates a local test command; it is not a bank receipt or evidence of an independent validator organization.

Pinned main's protected capability capsule accepts a closed public-source selection program and native approved research work. Arbitrary numerical components are outside that admitted scope. The study therefore reports its numerical component as `NOT_QUALIFIED_UNSUPPORTED_SCIENTIFIC_SCOPE`. It does not encode a numerical basis as a different supported skill to bypass admission. Existing ENS reader freshness, direct-label authority, revocation and settlement requirements remain unchanged.

Local process transfer and native local authentication are both executed, but they are different observations. The recipient loads only a verified basis and fresh numerical inputs; it does not acquire proprietary model weights or training snapshots. There is no live authenticated `*.agent.agi.eth` network in this experiment. Deploying a numerical service to all currently authorized agents remains an integration and qualification task.

The foundation reading includes Transformational Intelligence; the combined Gym, Specialist ASI and Successor Ω work; the Validated Skill Graph; and Proof Loops [12, 13, 14, 15]. Their architectures and synthetic constructions guide the protocol but are not treated as measured commercial outcomes. The companion records exact original source hashes and reading coverage.

5 Executed technical experiment

5.1 A useful quantity and a checkable bound

For a real, stable dynamical model with stochastic input, the stationary covariance X satisfies

$$AX + XA^\top + BB^\top = 0. \tag{2}$$

The synthetic family has 192 states and $A = -D + K$, with positive diagonal damping D and skew-symmetric coupling K. Two localized columns of B describe disturbance channels. Parameters vary damping, coupling, heterogeneity and input scale. All values are nondimensional research inputs; no physical plant was fitted.

For $dx = Ax\,dt + B\,dW$ with independent unit-intensity white-noise channels, zero direct feedthrough and $C = I$, $\mathrm{tr}(X)$ is the stationary expected squared state norm and the squared $\mathcal{H}_2$ norm. It is not the $\mathcal{H}_2$ norm itself; different disturbance normalization or output weights change the quantity being sold.

Let $\mathrm{sym}(A) \preceq -\alpha I$ for a verified $\alpha > 0$. For an approximate covariance $\widehat{X}$, form the full residual

$$R = A\widehat{X} + \widehat{X}A^\top + BB^\top. \tag{3}$$

The contraction bound $\|e^{At}\|_2 \le e^{-\alpha t}$ and the integral error representation yield

$$\|X - \widehat{X}\|_F \le F = \frac{\|R\|_F}{2\alpha}, \qquad |\,\mathrm{tr}(X - \widehat{X})| \le \delta = \sqrt{n}F. \tag{4}$$

When $\mathrm{tr}(\widehat{X}) - \delta > 0$, a relative trace-error bound is

$$\epsilon_{\mathrm{cert}} = \frac{\delta}{\mathrm{tr}(\widehat{X}) - \delta}. \tag{5}$$

Otherwise the solver refuses a finite relative certificate. For output matrix C, the corresponding energy error is bounded by $\|C^\top C\|_F F$. We use $C = I$ and the trace objective. This is a standard consequence of dissipativity, not a new theorem. Stability alone does not justify this particular bound.

The implementation validates matrix dimensions, finite values, covariance symmetry and the dissipativity claim. A Gershgorin sufficient condition is used where applicable, with an eigenvalue check otherwise. Basis dimensions, orthogonality and exact file hashes are checked before use. The bound is evaluated in floating point; no interval-arithmetic enclosure of rounding error is claimed. It certifies numerical solution error under stated matrix assumptions, not model-form error or physical robustness.

5.2 What is learned

Six originating solutions form an aggregate covariance. Its leading eigenvectors define an orthonormal reduced state basis of rank 24 or 48. Given basis V, a new query solves

$$(V^\top AV)Y + Y(V^\top AV)^\top + (V^\top B)(V^\top B)^\top = 0, \qquad \widehat{X} = VYV^\top. \tag{6}$$

The resulting array is a data-dependent executable artifact: changing the examples changes the representation. This is actual numerical learning through work examples. It is not frontier-model weight training or a claim of invented POD theory.

Optional Gym practice evaluates the rank-48 work basis on 20 development cases, selects the four largest residual certificates, pays for four full solutions and refits at the same rank. Equal rank isolates the effect of additional examples from simply allocating more representation capacity. Rank 24 is a separate compression ablation. No protected confirmation case is used to select training examples.

The ordinary alternatives are SciPy's Bartels–Stewart solve, adaptive block Krylov with reorthogonalization and residual stopping, and a fixed central-instance rank-48 Krylov basis. The adaptive method builds a query-specific basis and checks its certificate every eight dimensions up to rank 96. The fixed basis is an ablation, not a representation of every conventional library. Optimally configured rational/extended Krylov or production M.E.S.S. methods could be stronger still; their availability makes a broad superiority claim less defensible.

5.3 Freeze, splits and corrections

The initial protocol fixes six origin examples, 20 development cases, 64 protected confirmation cases, 32 shifted cases and 32 early prototype cases. Seeds, parameter boxes, rank, learning budget, error requirements and deadlines are serialized before confirmation. The original implementation and all arrays remain byte-identical in an archive. The protected quality gate requires a relative trace certificate at most 1%, an ex-post trace error at most 1%, and an ex-post Frobenius error at most 2%; the empirical pass threshold is 95%. Standard and interactive 32-case deadlines are respectively five seconds and 0.25 seconds.

Initial code trusted a caller-provided damping bound and marked the direct reference accepted without applying the common certificate gate. Review corrected both boundary defects. Scoped revalidation preserved every initial acceptance count and all four basis hashes. These revalidated cases are explicitly exposed cases, not a second fresh confirmation. The final prototype uses a new seed after freezing both corrected implementations and the unchanged proposal.

Each measured method includes matrix assembly, its numerical solve and full residual certification. Exact-reference error calculations are ex-post research audits and are excluded from serving time, while their cost belongs in the study's validation budget. Production fallback receives only the deployable certificate, not the hidden oracle. Inspection confirmed that the deployable certificate and the initial audit gate made the same decision on every initial row.

6 Experiment A: a speed advantage without a robust frontier crossing

Method	Protected in-range	Shifted family
Direct Schur	64/64	32/32
Adaptive Krylov, before fallback	64/64	22/32
Fixed central Krylov48	2/64	0/32
Work basis24	17/64	0/32
Work basis48	62/64	0/32
Gym basis48	62/64	0/32

Table 3: Actual initial numerical certification outcomes. Every refusal and failed ablation is retained.

The work basis learns substantial structure relative to a fixed central basis. Rank 48 is materially more reliable than rank 24. The Gym's four additional examples change the basis and improve some residual

magnitudes, but do not increase the protected acceptance count. All retained bases fail certification when damping is reduced and coupling increased. Numerically plausible answers are withheld when the bound fails.

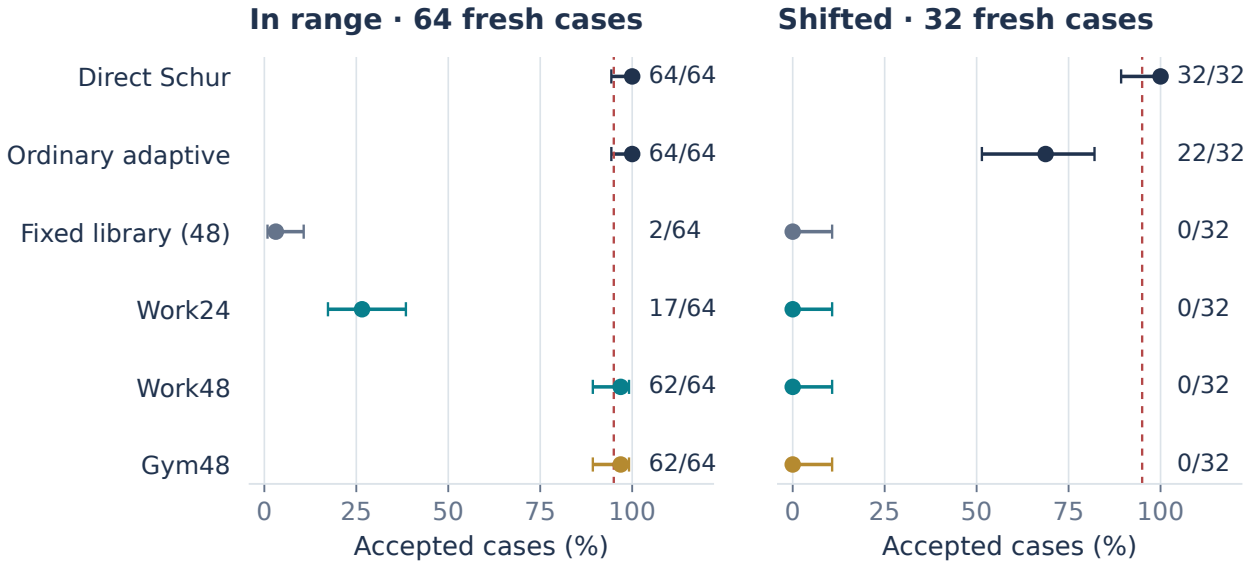


Figure 1: Numerical acceptance on the protected generator and the declared distribution shift. Sampling intervals describe the synthetic generator; they do not certify a population service-level agreement or physical reliability.

Initial standard 32-case batches take 1.10–1.15 seconds for Schur, 0.177–0.224 seconds for adaptive Krylov and 0.078–0.081 seconds for Gym48. All relevant methods fit the standard five-second numerical contract. The ordinary adaptive method also fits the 0.25-second threshold in the original two batches. Thus a comparison only against the dense reference would falsely suggest unique enablement.

After input and certificate checks are hardened, ordinary adaptive batch times straddle 0.25 seconds. The full fallback-chain revalidation yields 0.2429 and 0.2507 seconds for ordinary delivery, 0.1138 and 0.1174 seconds for work reuse, and 0.0918 and 0.1296 seconds for Gym reuse. These are useful measurements but insufficient evidence of a robust latency boundary: source overhead, host variation and the small number of batches matter. The protocol threshold is not moved to favor either result.

Fallback restores numerical acceptance on every tested case. In-range, each complete policy accepts all 64 revalidated instances. Under shift, each accepts all 32, but ordinary delivery takes 1.677 seconds per batch, work-first fallback 1.746 seconds and Gym-first fallback 1.801 seconds. Attempting an unsuitable specialist adds cost. A scope guard should route clearly shifted inputs to an appropriate ordinary method; the forced-use treatments retain this cost to expose the failure.

7 Experiment B: the next useful prototype

The adjacent mission selects added damping $g \in \{0, 0.1, \ldots, 0.7\}$ for each of eight fresh systems. It preserves the matrix family and minimizes

$$J(g) = \operatorname{tr}(X_g) + 0.15g^2, \qquad A_g = A - gI. \tag{7}$$

The coefficient, grid and two-second complete eight-case numerical deadline are explicit research-service hypotheses. This kernel deadline includes certification, fallback and ranking refinement, but excludes

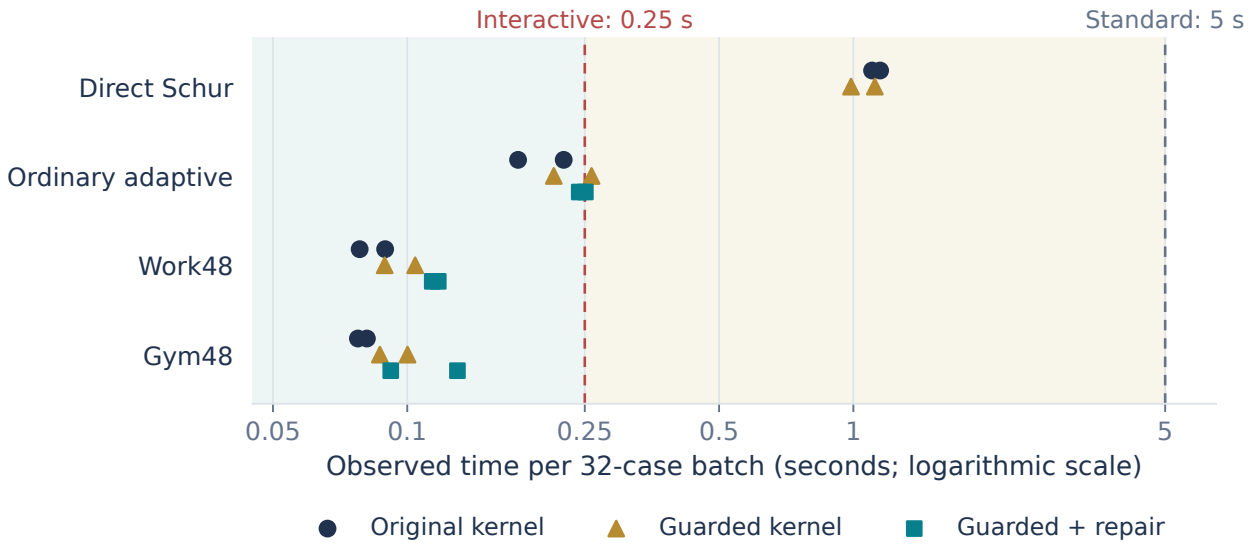


Figure 2: Frozen standard and interactive boundaries, with the original observations and exposed-case guarded measurements distinguished. A threshold-sensitive observation is not a dependable commercial SLA.

the modeled frontier workflow's preparation and packaging time. No claim of continuous or global optimality is made. A numerical certificate gives an interval for each objective. Candidates whose lower bound exceeds the smallest upper bound are eliminated. Ambiguous possible winners receive ordinary direct refinement; unresolved overlap requires abstention.

Both enterprises receive the same customer-evidence hypothesis, candidate grammar, objective and ordinary tools. One also receives the work-derived or Gym-derived basis. Their proposals are frozen before final fresh confirmation. The formal proposal catalogue was serialized after the exploratory prototype, so the study does not claim that catalogue was pre-registered before exploratory seed 6401. The original numerical criteria were frozen before that run. After safety corrections, one additional fresh seed, 7401, confirms the unchanged proposal and corrected source without retraining or threshold changes.

Final fresh policy	Seconds	Optimal / 8	Direct refinements	Meets 2 s
Schur	2.0554	8	0	Observed miss
Adaptive with fallback	0.4662	8	2	Yes
Work48 with fallback	0.1728	8	0	Yes
Gym48 with fallback	0.4592	8	8	Yes

Table 4: Fresh seed7401. Every method has zero objective regret against the independently computed discrete-grid oracle. The small direct-solver miss is timing-sensitive and is not a robust impossibility claim.

The work basis saves 0.293415 seconds per complete eight-case service, a 62.9% reduction relative to the ordinary adaptive comparator. Gym saves only 0.007064 seconds on this fresh service. On the exposed exploratory prototype, Gym is slower than the ordinary comparator. The difference follows actual refinement work, not an assigned performance multiplier.

The result is an executable adjacent engineering service that benefits from reuse, while ordinary Beta already qualifies. It is not a specialist-enabled new market. The finite proposal grammar does not establish that a frontier model independently invented the offering. Usefulness is grounded in an

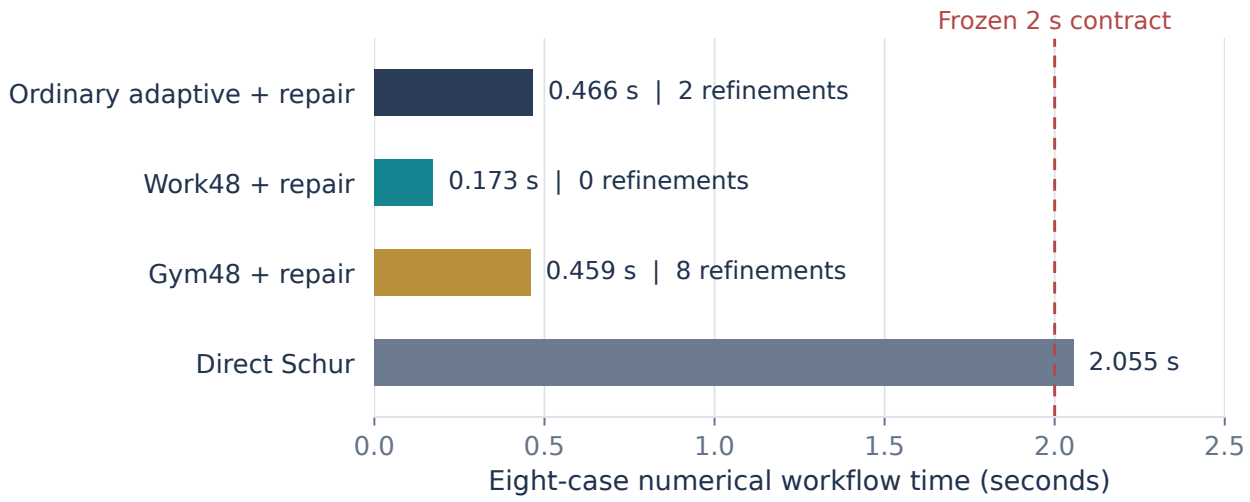


Figure 3: Complete numerical-prototype time includes failed certificates, ordinary fallback and ranking refinement, but excludes assumed frontier-model preparation. A faster primitive can lose its advantage when composed into a verified workflow.

established computation, but buyer demand and price remain unvalidated. Eight successful selection cases also provide limited reliability evidence; they do not establish dependable unfamiliar-domain superiority.

8 Transformational intelligence as a testable intervention

The transformation considered here changes the representation and work allocation while preserving the mission. The obstacle is repeated full-state covariance computation. The proposed change is to reuse an empirically learned subspace, certify the result with an independent residual, and invoke ordinary computation when the certificate or ranking is inconclusive. The decisive test is a fresh complete service against an unrestricted ordinary alternative.

Three lessons follow. First, the representation really transfers: a separate process loads a digest-verified basis and computes fresh outputs without the training corpus. Second, the learned representation changes which internal computation is needed, but not which tested offering is available to a competent ordinary enterprise. Third, more practice is not necessarily better preparation for the next mission. Ranking uncertainty, rather than covariance approximation alone, determines downstream refinement cost.

The experience can inform a better decision: retain the work basis as a research candidate, avoid automatic Gym expenditure, require a scoped residual check, and look for a task whose economically important bottleneck is actually changed. This edition does not measure a general improvement in an agent's opportunity judgment. Its economic selector is a finite, inspectable decision rule using the measured evidence. A live learned opportunity-selection policy requires a separate held-out sequence of proposals and outcomes.

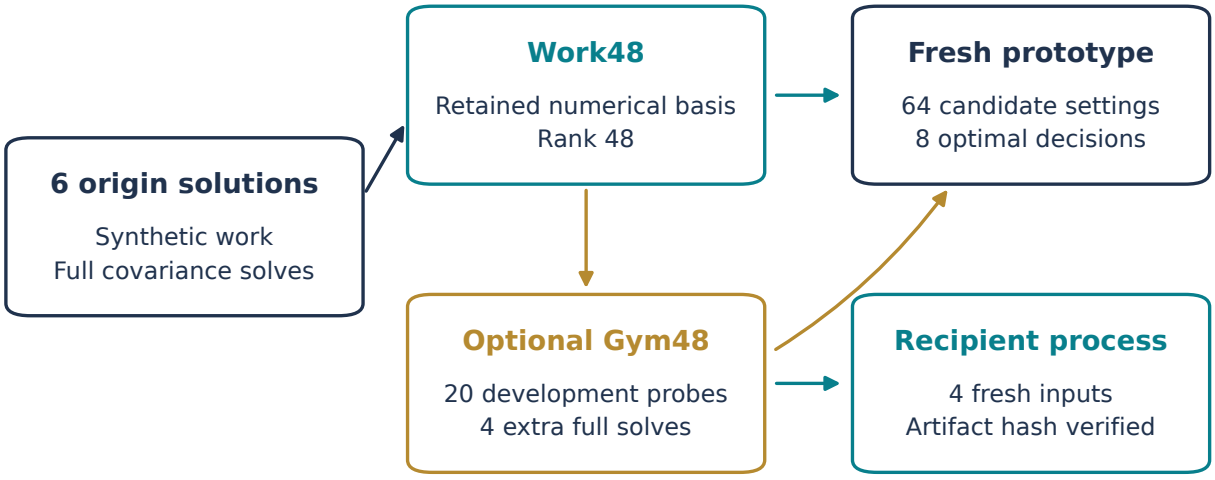


Figure 4: Capability lineage and evidence boundaries. Learning, local transfer, native accounting and production admission have separate statuses. An executed local artifact does not silently become an authorized network service.

9 Experiment C: complete conditional enterprise economics

9.1 What is observed, sourced and assumed

The numerical artifacts, accept/refuse outcomes, query timings, fallback calls and transfer tests are observed executions on synthetic systems. Current model identities and unit prices are sourced from official documentation. Token workloads, automated preparation latencies, customer demand, trust, workload, pricing references, acquisition costs and operational expenses are explicit assumptions. Buyer choices, collections and commercial failures are simulated. Confidence intervals for the simulated mean measure Monte Carlo estimation uncertainty under those assumptions; they do not validate the assumptions.

The model covers 60 operating days. Every matched policy receives the same USD2,500 seed capital, total daily processing capacity, ordinary tools, customer draws, candidate catalogue and price grid. The seed is capital, never revenue. Origin examples in this edition are unpaid research inputs: their compute and delivery-equivalent model budget are charged once in C, with zero collections. In a genuinely paid-origin implementation, the learning necessary for accepted delivery would belong in A and would not receive another research bill.

Buyers may choose AI delivery, human delivery, deferral or no purchase. Utilities depend on an explicitly assumed value distribution, trust-adjusted usefulness and price. The enterprise evaluates eight prices relative to the estimated human-service anchor; it can refuse incompatible inputs, negative expected contribution, insufficient funds or unavailable reserved capacity. It reserves capacity before realizing execution time or failure, preventing perfect foresight about which jobs will be cheap. Failed acceptance, refunds, collection delays and defaults remain in the records.

The five-second audit/batch gates and two-second selection gate measure numerical work. The

enterprise separately assumes 90, 120 and 180 seconds of frontier-model preparation and packaging, reserving 95, 125 and 182 seconds respectively. No live end-to-end SLA is established. Unused job reservations are not reclaimed during the day, so faster kernels do not automatically produce more accepted work; this conservative scheduling rule limits the model's throughput conclusions.

9.2 Optional actions and comparable alternatives

The competent ordinary baseline may pursue the same standard batch and finite-grid selection offerings. It can prototype, price, decline work and hold cash. Private-work, shared-work and shared-Gym treatments force the specified capability investment to isolate its consequences. Private duplication is an explicit ablation with repeated integration/assessment, not a universal claim that every private specialist architecture must pay four times as much. An ordinary-library arm has identical contents, permissions and costs to shared work; no graph-specific benefit is assumed.

The selective policy pays for assessment and compares retention, Gym and holding cash. Its capability screen uses an optimistic forecast of the largest measured per-order compute saving, observed demand information and remaining horizon. Future upkeep affects valuation but is charged only when incurred. This is a bounded screening rule, not a global solution of enterprise pricing, scheduling and research. Its optimism makes rejection informative when even the favorable repayment estimate is too small.

Prototype commercialization has an assumed success probability of 0.85, separate from numerical correctness. Each candidate gets one attempt. Failed prototypes keep their charges and consume preparation capacity. Integration, Gym and common production setup also displace processing capacity. Opportunity cost appears as displaced orders and paired enterprise outcomes; it is not subtracted a second time as a cash expense. A separate growth arm may purchase extra capacity from available operating surplus, with the same option for every policy. Scale growth by itself is not evidence of learning-driven compounding.

9.3 A ledger that does not invent revenue

Costs are disjoint: A contains modeled customer delivery, intake, corrections, refunds and allocated operations; B contains additional assessment, retention, admission, transfer preparation and upkeep; C contains unpaid origin research, optional Gym, productization and common setup. The native replay assigns unique identities to 27 bucket/category aggregates; original activity records remain in the exemplar. The model assumes USD8 per acquired qualified lead and USD12 per operating day for allocated monitoring, support and administration. These amounts are hypotheses, not observed invoices. There are no arbitrary routine human-review minutes; the proposed product's scope excludes physical commissioning and professional sign-off.

Let K be seed capital, R_t actual simulated collections received by t, C_t accrued modeled costs, P_t paid modeled costs, and Z_t reserves. The reported quantities are

$$\Pi_t = R_t - C_t, \qquad \text{Cash}_t = K + R_t - P_t, \tag{8}$$

$$\text{Obligations}_t = C_t - P_t, \qquad \text{Available}_t = \text{Cash}_t - \text{Obligations}_t - Z_t. \tag{9}$$

This is a conservative collected-basis operating surplus, not an assertion of audited accrual-accounting profit. Gross unpaid invoices are reported separately. Refunds are explicit costs and are paid immediately when a failed order's collection arrives. Customer savings, token issuance, speculative opportunity value and uncollected invoices are not earned revenue. Applicable $AGIALPHA mechanisms are outside this operating ledger.

The native ledger conformance fixture uses deliberately fictional amounts and is kept separate from the enterprise results. Unknown costs remain unknown, and native completeness gates prevent a missing cost from becoming zero. An aggregate replay of a recorded enterprise history checks that model totals and the unchanged native ledger agree. This proves accounting behavior for supplied attestations, not the truth of customer or provider payments.

9.4 Results and where investment stops paying

The frozen confirmation executes 256 paired seeds, 12 scenarios and six policies (18,432 enterprise histories). Table 5 reports central assumed-market outcomes over 60 operating days. These amounts are entirely simulated USD, not earned customer revenue. All optional capability treatments remain below their competent ordinary alternative in this experiment.

Policy	Mean surplus	Paired increment	Available funds
Competent ordinary Beta	51,655.06	0.00	54,055.06
Private work (forced)	51,495.06	-160.00	53,895.06
Shared work (forced)	51,594.06	-61.00	53,994.06
Shared Gym (forced)	51,569.06	-86.00	53,969.06
Ordinary library (matched)	51,594.06	-61.00	53,994.06
Selective assessment	51,647.06	-8.00	54,047.06

Table 5: Conditional simulation. Surplus equals simulated collections less all accrued modeled costs. Seed capital is excluded from surplus and included only in funds. Forced treatments isolate capability investment; the library has identical contents and permissions to shared work.

Ordinary Beta's mean collected-basis surplus is USD 51,655.06, with a Monte Carlo 95% interval for the mean of [50,115.40, 53,194.72]. The fifth and ninety-fifth percentiles of individual simulated outcomes are USD 30,532.70 and USD 65,390.86. Mean simulated collections are USD 59,245.96; mean A/B/C costs are USD 7,326.64, USD 0.00 and USD 264.26. The mean interval measures Monte Carlo estimation uncertainty under fixed assumptions, not uncertainty in actual demand or prices.

The selective policy adopts an additional specialist in 0 of 3,072 scenario-seed histories. Its central paired increment is minus USD8, the retained assessment cost. The shared-work treatment loses approximately USD61 despite positive absolute delivery surplus. This separates the commercial foundation from the additional learning investment.

The corrected low-willingness-to-pay policy stops both lead acquisition and optional prototypes. It still loses USD 874.26 because the fixed 60-day operating commitment and common setup/origin research costs remain. The preserved initial policy lost USD 984.26 in that scenario, including USD 110 of avoidable prototypes. One declared amendment conditioned productization on the marginal contribution of an acquisition-viable portfolio; a new 256-seed cohort confirmed the correction without changing technical criteria, prices or demand assumptions. This is a researcher-corrected policy, not evidence of autonomously learned commercial judgment. The growth arm can earn much more under its assumed high demand because every policy may purchase extra capacity. That result concerns scalable ordinary delivery under the assumed market, not a learned expansion of what is technically possible. Quality-adjusted units use declared weights 1/3/6 for accepted output, and a separate metric divides them by processing hours. These are accounting weights, not measured buyer value.

The decisive mechanism is the size of the absolute useful saving. At a hypothetical USD0.50 per core-hour, the final fresh work-basis saving is approximately USD0.00004075 per eight-case service. Even USD20 of incremental integration requires about 490,772 such services to repay through compute

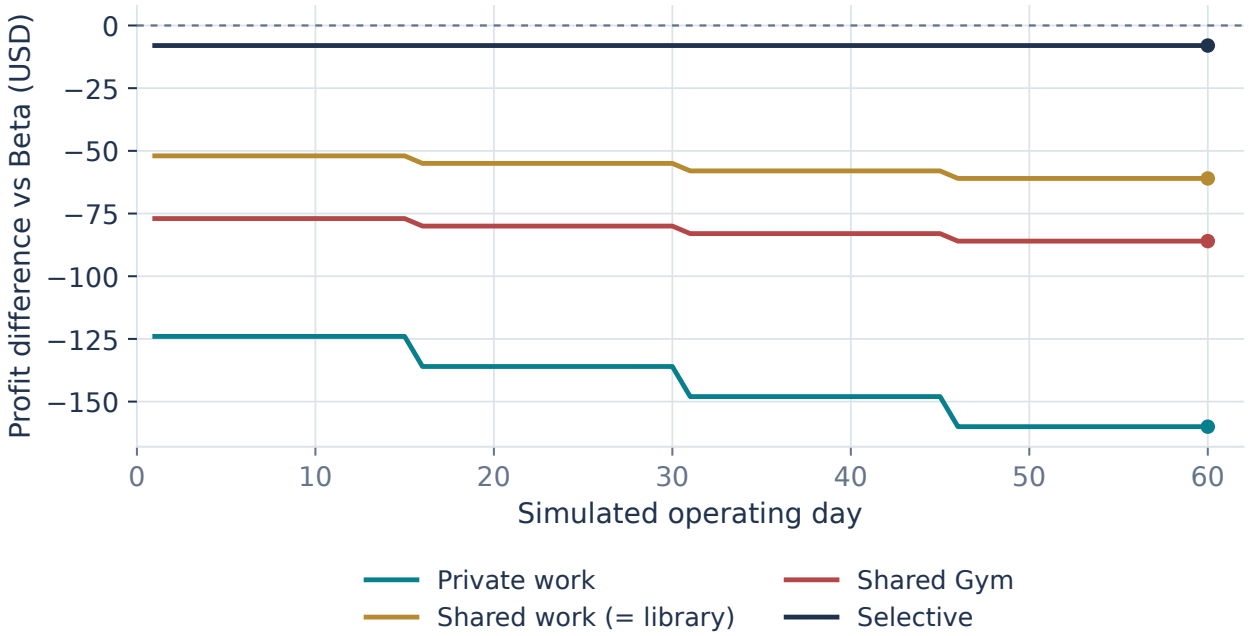


Figure 5: Conditional cumulative incremental surplus relative to equally resourced ordinary delivery. The negative lines retain additional assessment, integration and upkeep; no customer premium is assigned to the specialist.

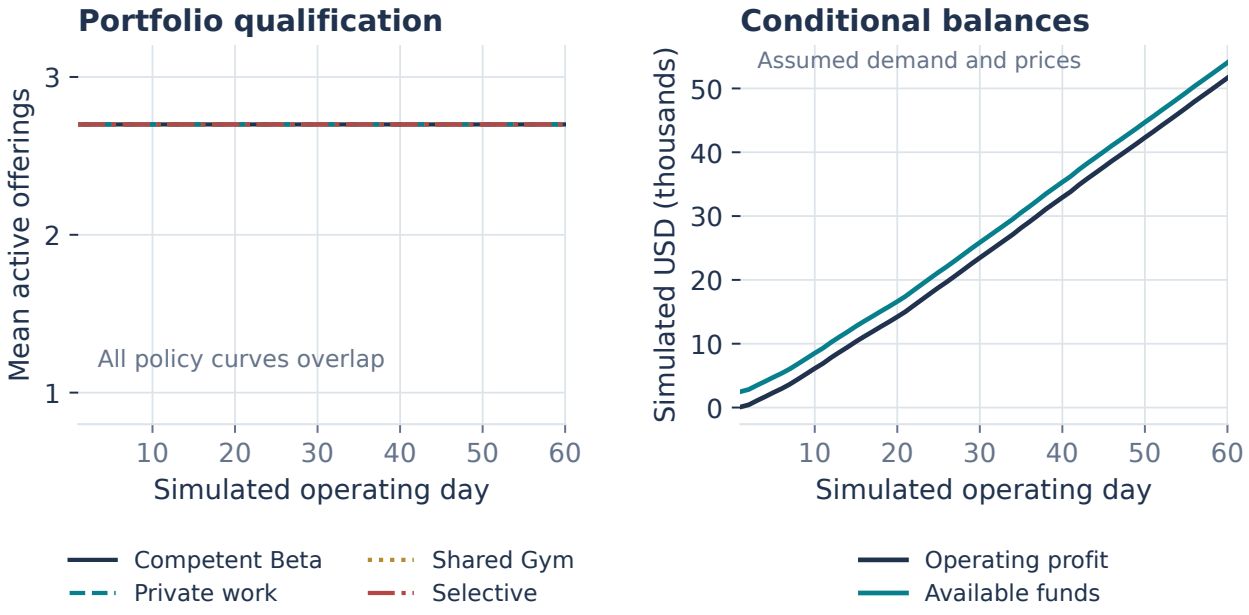


Figure 6: Conditional portfolio and available-funds paths. Left: all policies qualify the same offerings. Right: selective-policy surplus and available funds. Productization can add offerings for both enterprises; this is not specialist-created feasibility.

Scenario	Beta surplus	Shared work increment	Selective increment
Base	51,655.06	-61.00	-8.00
Low demand	3,528.12	-61.00	-8.00
Low wtp	-874.26	-61.00	-8.00
High failures	29,669.29	-61.00	-8.00
High integration	51,655.06	-349.00	-8.00
Poor transfer	51,655.06	-61.00	-8.00
Capacity pressure	173,569.36	-60.99	-8.00
Regime shift	51,655.05	-61.00	-8.00
Slow collections	23,285.99	-61.00	-8.00
Rapid frontier progress	51,698.44	-61.00	-8.00
High compute price	51,654.54	-60.78	-8.00
Growth financed by surplus	787,488.94	-60.97	-8.00

Table 6: Stress outcomes under deliberately changed assumptions; all values are simulated USD. Growth purchases additional capacity from surplus and is separately identified. Larger output from that capacity is not evidence of Alpha compounding.

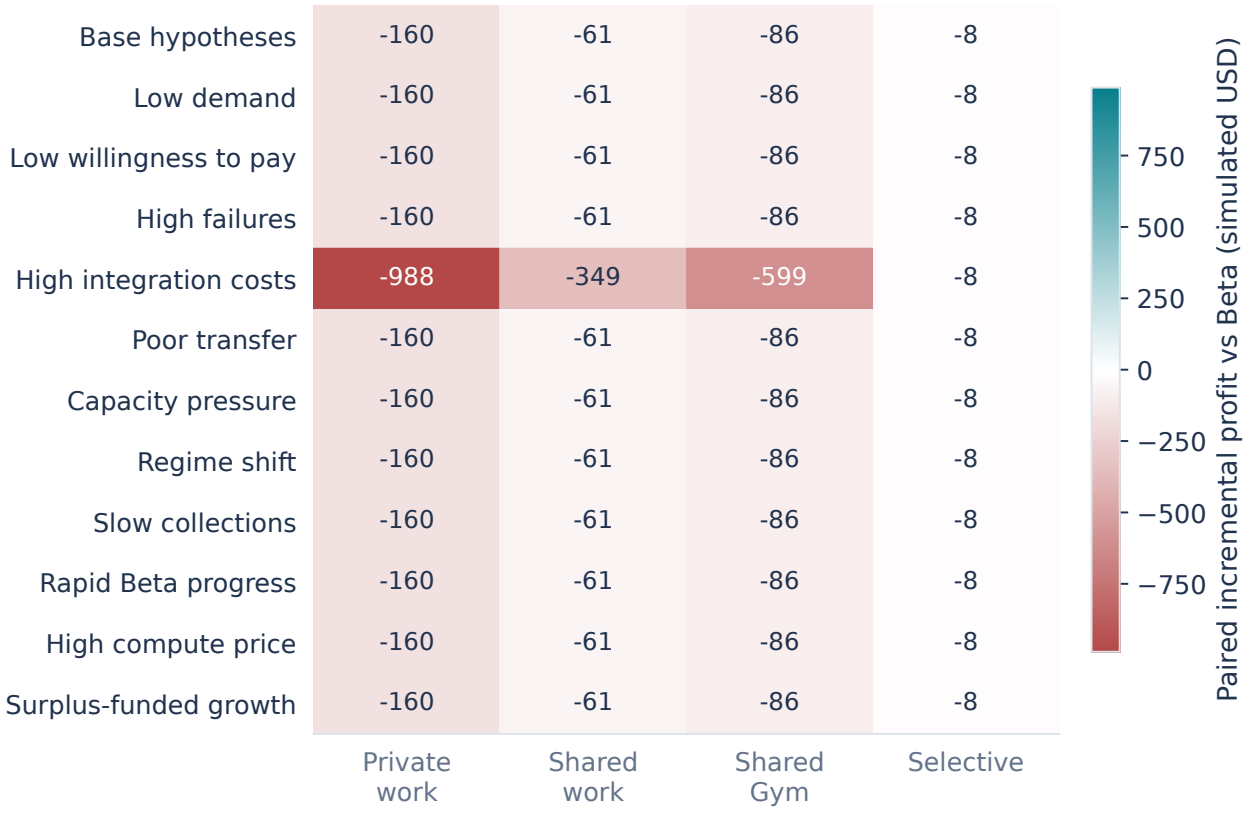


Figure 7: Paired policy increments across assumption stress tests. The market and operating assumptions, not Monte Carlo replication, determine whether a scenario represents a real opportunity.

alone. A USD61 retention, sharing and upkeep allowance requires about 1.50 million services, before demand risk or obsolescence. These are conditional arithmetic boundaries, not demand forecasts.

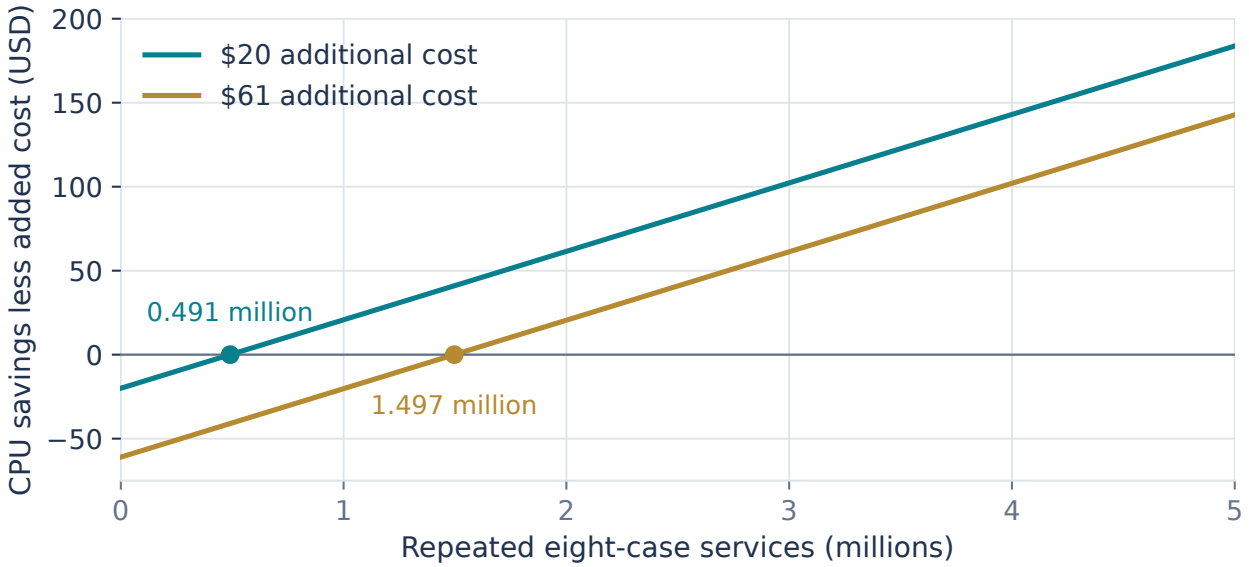


Figure 8: CPU-only repayment sensitivity for the measured fresh service saving. Changing workload or hosting price changes the arithmetic; it does not establish customer demand or an independently valuable deadline crossing.

The study does not assign a premium or additional demand merely because a specialist is faster. A customer may value a sufficiently responsive service, but that value must be established independently. Here the ordinary method already satisfies the final prototype contract, while the stricter interactive boundary lacks dependable evidence. Thus the modeled economic case for additional specialization rests on tiny resource savings and fails at the tested workloads.

The stress suite varies demand, willingness to pay, packaging failure, integration cost, transfer reliability, capacity pressure, regime shift, collection delay, common technology progress and hosting price. Progress is modeled as a common cost/latency sensitivity, including fallback for every arm. It is not a measured comparison with a newer frontier model. The ordinary library and shared-work arms remain identical by construction, so graph-specific causation is neither inferred nor claimed.

10 Current frontier-model operating blueprint

Official documentation was checked on 27 October 2026. The intended operating roles can be assigned to the following verified API identifiers [7, 8, 9, 10]:

The pages describe 1,050,000-token context and up to 128,000 output tokens. Astra and Sol expose reasoning settings from low through max; Terra additionally supports none. Tool calling uses the Responses API. Fine-tuning is not supported for these listed models. Public documentation does not establish this account's API entitlement, quota or billing. ChatGPT Pro is a plan, not a model identifier or automatic API access [11].

The modeled per-order frontier workflow allocates appropriate reasoning, coding, validation and computer-use roles and an assumed USD0.71 text-token budget, plus separate tool and operational allowances. The budget is an explicit workload assumption, not a trace of live calls. No quality

Model	Intended role	Input	Cached	Output
`gpt-6-astra`	Scientific reasoning, strategy, validation	10.00	1.00	50.00
`gpt-6.1-sol`	Coding, tools and professional execution	2.00	0.10	10.00
`gpt-6.1-terra`	Visual reasoning and precise computer use	1.00	0.05	5.00

Table 7: Standard USD per million text tokens, as documented on the inspection date. Long-context surcharges, speed modes, tools and hosting are additional considerations. These are sourced prices, not measured study consumption.

or speed result in this paper is attributed to actual Astra, Sol or Terra execution. The executable numerical baseline represents competent ordinary tools available to such an operator; it does not measure the capability ceiling of frontier reasoning.

A live evaluation should give both arms the same objectives, context, ordinary libraries, revision allowance, tool access and resource budget. Model upgrades must be introduced symmetrically and tested on new missions. Faster models, more compute, better prompting and specialist learning must be separately attributable. The present common-progress sensitivity is useful for economics but does not replace that experiment.

11 Operator decisions: offer, qualify, price and stop

Offer first. Begin with the standard offline numerical audit of already validated, rights-cleared dissipative models. Require a machine-readable input contract and a named useful decision. Produce reproducible outputs and numerical-error evidence. Refuse unsupported dynamics, absent units, unknown rights or requests for physical safety sign-off. The first live pilot should determine whether this narrowly scoped deliverable saves a real team's work.

Price after scope qualification. Use estimated human-equivalent effort as a transparent reference where the scope matches, then test willingness to pay. In the central assumed market, the finite search selects USD165, USD330 and USD660 for the audit, standard batch and finite-grid service. Those are scenario outputs, not validated market-clearing prices or quotes to a customer. Record acquisition, preparation, corrections, support, obligations and collections before calling an offering profitable.

Choose the capability with a repayment mechanism. The work basis is the better measured candidate. It should not receive automatic production admission or extra Gym funding. For the tested service, its compute saving is too small to pay normal additional integration and upkeep at the modeled workload. Retain it as inspectable research evidence, and reconsider only when an independent contract, larger measured workload or different expensive bottleneck provides a credible mechanism.

Treat the next offering as an experiment. The finite-grid damping-selection prototype works and transfers, but ordinary tools already deliver it. Its commercialization may still be worthwhile as an ordinary service. That would validate product development, not specialist-created frontier expansion. The continuous-monitoring proposal remains rejected until its missing prototype, model assurance and buyer evidence are supplied.

Stop or redirect. A failed certificate requires ordinary fallback or refusal. Scope shift should bypass an unsuitable basis. Gym should stop when fresh service-level benefit is absent. Productization should stop when expected contribution does not cover its cost, and acquisition should stop when the assumed market cannot cover acquisition and intake. Available funds, not gross receipts or speculative option

value, constrain additional commitments.

12 Limitations and the next decisive experiment

The result has a narrow technical domain. The matrices are synthetic and dissipative by construction; dimensionality, topology, input locations and parameter ranges are limited. Learning examples and confirmation instances come from a known generator. The shifted family tests one failure mode, not all unfamiliar systems. No physical model, laboratory experiment, industrial controller or human buyer was validated.

The timing sample is small, one host is used, execution order is not a full randomized crossover, and complete interactive reliability is not established. Exact source identity and separate initial/revalidated/fresh datasets preserve what was observed, but do not create stronger generalization. The fresh selection cohort contains only eight systems. Measured numerical bounds do not provide interval-certified floating-point proofs.

The opportunity search is finite and researcher-specified. It does not establish open-ended autonomous invention or independent demand discovery. Neither does the shared-artifact experiment establish that every recipient acquires a specialist's general intelligence. A numerical basis is a compatible reusable component; a service would instead provide bounded access to computation. Graph topology is not tested as a source of advantage.

Commercial results are entirely conditional. Token counts, costs, buyer utilities, workload, packaging risk and capacity are assumed. The model omits some possible strategies and cannot be called a globally optimal enterprise controller. It assumes a fixed operating horizon with allocated overhead; shutting down the enterprise or renegotiating obligations may be a better alternative in an adverse market. No simulation confidence interval validates a price anchor or demand curve. No actual collections were earned by this study.

The next decisive experiment should begin with a real beneficiary and a demanding fixed contract for which the strongest ordinary approach is close to an economically important limit. It should include paid or otherwise independently validated demand, rights-cleared originating experience, a properly costed specialist campaign, a fresh recipient, and a next-offering comparison that permits ordinary acquisition alternatives. Its strongest conclusion might again be to decline training. That decision is productive when it preserves resources for a better-supported opportunity.

13 Reproducibility, publication and disclosures

The companion contains original protocols, source freezes, raw per-case CSVs, parameter manifests, learned arrays, archived pre-correction source, final fresh prototype evidence, economic configuration and paired histories, native accounting attestations, source notes, reviewer records, figures, a bilingual browser edition and rebuild instructions. Results identify their exact artifact and source version. Reproduction writes to a new directory so original observations remain intact.

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Conclusion. Specialist learning produced a real transferable numerical advantage. The tested next service was already available to a competent ordinary enterprise, and extra practice did not create a better complete service. Profitable ordinary delivery remains a valid foundation wherever the market assumptions hold. The defensible policy is to invest only when a capability changes a valuable constraint or repays its complete additional burden, and to stop when the evidence supports ordinary delivery instead.

A Evidence and accounting map

Artifact / activity	Evidence classification	Accounting treatment
Six origin full covariances	Executed unpaid research examples	C once in this edition; A only in a genuinely paid-origin deployment
Work basis fit and admission assessment	Executed fit; operational cost assumed	Additional B; no second origin solve charge
Twenty Gym assessments, four solves and refit	Executed technical learning	Optional C
Independent oracles, safety revalidation and fresh confirmation	Executed research validation	Retained research/assessment budget, not free serving performance
New offering packaging and failed candidates	Numerical prototype observed; commercialization simulated	C, including failed attempts and capacity use
Customer jobs, refunds and corrections	Simulated market and delivery packaging	A with actual modeled collections separated
Rights/access preparation, recipient integration and upkeep	Hypothetical operating work; local transfer observed	B, with no live ENS claim
Seed, reserves and invoice receivables	Simulated financial state	Never earned revenue; obligations shown separately

Table 8: A complete modeled activity budget is not a measurement of actual commercial expenditure. Missing live quantities remain an explicit deployment limitation.

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